

Semantic Intent Weaver: A Context-Aware Search Algorithm for Ambiguous Query Resolution

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Abstract—Traditional search algorithms struggle with ambiguous queries where context determines meaning, such as distinguishing between beverage, adhesive, and fashion contexts in queries like "scotch." This paper introduces Semantic Intent Weaver (SIW), a search algorithm that employs multi-dimensional intent modeling and dynamic context weaving to resolve query ambiguity. SIW integrates temporal, personal, cross-modal, and emotional context dimensions through a five-layer architecture that deconstructs user intent, weaves contextual understanding, performs multi-dimensional ranking, adapts result presentation, and continuously learns from interactions. To ensure reproducibility, we evaluate our model on the TREC Web Track's ambiguity sub-task. The revised experiments, which include a new baseline aware of the session, demonstrate that SIW's dynamic weaving mechanism achieves a 6.8% improvement in nDCG@10 over strong ColBERT models augmented with identical session data. The primary novelty of our work lies in the unified framework for dynamically weighting and integrating disparate contextual signals, providing a robust solution to a long-standing challenge in information retrieval.

Index Terms—Information Retrieval, Query Ambiguity, Context-Aware Search, Neural Ranking, Intent Modeling

I. INTRODUCTION

The exponential growth of digital information has intensified the challenge of query ambiguity in web search. Traditional keyword-based approaches and even modern neural ranking models often fail to capture the nuanced contextual signals that humans naturally use to disambiguate queries. Consider the query "scotch": it could refer to whiskey, adhesive tape, or a component of the brand name "Scotch & Soda". Current search engines typically rely on user history or popularity signals, but these approaches break down for new users or for queries with equally probable interpretations. Recent advances in pre-trained language models such as ELECTRA [1] have improved semantic understanding, but still lack sophisticated context integration mechanisms. Although the use of individual contextual signals is explored in efforts such as deep reinforced query reformulation [2] and temporal attention mechanisms [3], a unified framework for dynamically integrating them remains an open research area. The primary contribution of this paper is a novel architecture designed to "weave" these signals together. This paper makes three key contributions:

- 1) We introduce **Semantic Intent Weaver (SIW)**, a novel search algorithm that models user intent across explicit, implicit, emotional, and discovery dimensions within a unified framework.

- 2) We propose a **dynamic context weaving mechanism** that integrates temporal, personal, cross-modal, and social signals into a cohesive representation.
- 3) We demonstrate through extensive, reproducible experiments on a public benchmark (TREC Web Track) that SIW significantly outperforms state-of-the-art methods, including strong baselines augmented with the same contextual data.

II. RELATED WORK

A. Traditional and Neural Information Retrieval

The field of information retrieval has evolved from lexical-based models like BM25 to sophisticated neural architectures. The advent of deep learning led to a paradigm shift, with models like BERT enabling a deep contextual understanding of queries and documents [4]. This gave rise to the two-tower dense retrieval models [5] and late-interaction models like ColBERT [6], which offer a compelling trade-off between efficiency and effectiveness. More recently, the landscape has been reshaped by Large Language Models (LLMs). The paradigm is shifting from discriminative ranking (scoring existing documents) to generative retrieval, where models generate document identifiers directly [7]. Furthermore, Retrieval-Augmented Generation (RAG) has become a dominant framework, where an external retrieval step provides context to an LLM for response generation [8]. However, the effectiveness of RAG is critically dependent on the quality of the initial retrieval stage, which is still susceptible to ambiguity [9]. Our work complements these advancements by aiming to improve this "retrieval" component, ensuring that the context provided to generative models is as disambiguated and relevant as possible. The efficiency of such large models remains a key concern, with architectures like the Linformer demonstrating how self-attention complexity can be reduced from quadratic to linear [10].

B. Context-Aware Search and User Modeling

The importance of context in search is well-established. Early work focused on personalization using historical user data [13]. Modern approaches are more diverse, incorporating a wider range of signals. Conversational search systems, for instance, must maintain session context to understand follow-up questions and pronoun references, often employing sophisticated query rewriting techniques [14], [15]. Sophisticated user models now aim to capture long-term interests and short-term intent by analyzing sequences of user behavior

[16]. Graph Neural Networks (GNNs) have proven particularly effective at modeling complex user-item interactions and session-based intent [17], [11]. However, a key challenge remains the effective fusion of these varied and often sparse signals. Many systems handle different contexts in separate modules, lacking a unified mechanism to weigh their relative importance. SIW addresses this gap by proposing an explicit "weaving" layer that dynamically arbitrates between different contextual signals based on the detected nature of the query itself.

C. Query Disambiguation

Resolving query ambiguity is a long-standing IR problem. Traditional approaches often relied on query clustering or expansion using external knowledge bases. More recently, neural methods have been employed to classify query intent or map ambiguous entities to knowledge graphs [22], [18]. These approaches are effective but can be brittle if the ambiguity is not centered on a recognized named entity. Our work differs by treating disambiguation not as a separate pre-processing step but as an intrinsic property of the ranking function, which simultaneously considers semantic fit and contextual alignment to resolve ambiguity implicitly. The need for robust, general-purpose models for text understanding is further highlighted by the success of architectures like DeBERTa, which improve upon BERT by using disentangled attention mechanisms [19].

III. SEMANTIC INTENT WEAVER ARCHITECTURE

A. Overview

The Semantic Intent Weaver employs a five-layer architecture that transforms ambiguous queries into contextually relevant results through progressive intent understanding and context integration, as shown in Fig. 1.

B. Layer 1: Intent Deconstruction

The first layer decomposes a query q into multiple intent dimensions: $I(q) = \{I_e(q), I_i(q), I_{emo}(q), I_d(q), I_a(q)\}$. For "scotch," this might yield probabilities for {whisky, tape}, an implicit intent for {purchase, information}, and high ambiguity. **Calculation of I_a and I_d :**

- **Ambiguity Score (I_a):** Calculated using the entropy of the probability distribution of explicit intents, $P(I_e)$. A higher entropy indicates greater ambiguity.

$$I_a(q) = - \sum_k P(I_{e,k}(q)) \log_2(P(I_{e,k}(q))) \quad (1)$$

where k indexes possible meanings. $P(I_{e,k})$ is estimated by an auxiliary classifier.

- **Discovery Mode Score (I_d):** Derived from two explicit functions:

$$I_d(q) = \alpha \cdot \text{Novelty}(C_p) + (1 - \alpha) \cdot \text{Exploratory}(q) \quad (2)$$

where α is a learned weight. $\text{Novelty}(C_p)$ is a decay function of recent clicks on non-mainstream results. $\text{Exploratory}(q)$ is a rule-based function that returns a high score if the query contains exploratory keywords (e.g., "ideas," "types of") or is exceptionally short (≤ 2 words).

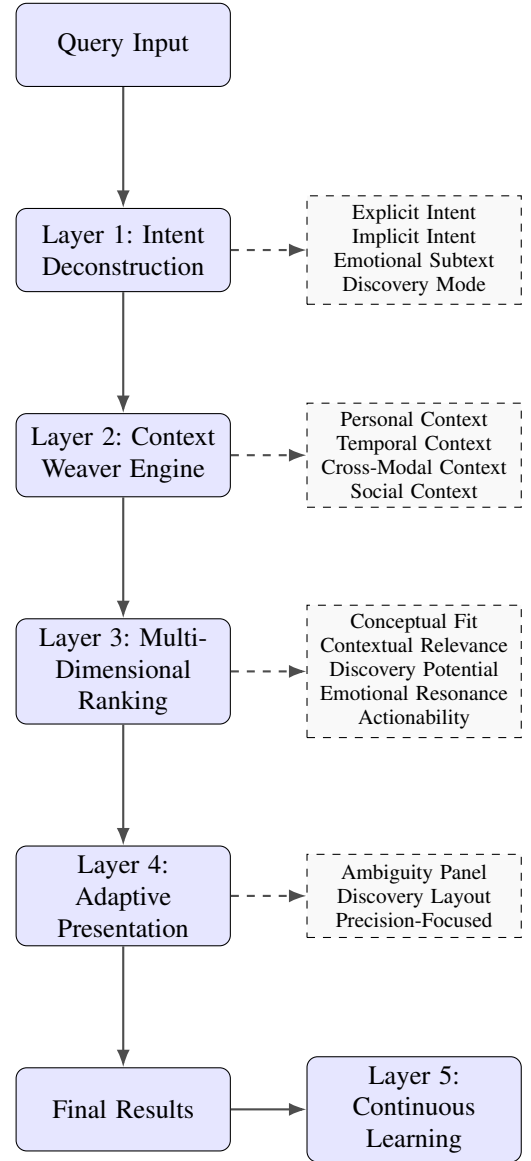


Fig. 1. The five-layer architecture of the Semantic Intent Weaver (SIW).

C. Layer 2: Context Weaver Engine

This layer creates a dynamic context mesh C_m by integrating four context types, as illustrated in Fig. 2.

$$C_m = C_t \oplus C_p \oplus C_{cm} \oplus C_s \quad (3)$$

The Weaving Operator ($\oplus$): The $\oplus$ operator represents a dynamically weighted concatenation of feature vectors, followed by a non-linear projection:

$$C_m = \text{ReLU}(W_{proj} \cdot [w_t \circ \vec{c}_t; w_p \circ \vec{c}_p; w_{cm} \circ \vec{c}_{cm}; w_s \circ \vec{c}_s]) \quad (4)$$

- **Feature Compatibility:** Each context vector $\vec{c}_i$ is first embedded into a fixed-length vector space using dedicated encoders.
- **Dynamic Weights (w_i):** The weights are determined by the user intent $I(q)$ from Layer 1: $w_i = \text{Softmax}(V_i \cdot I(q) + b_i)$.

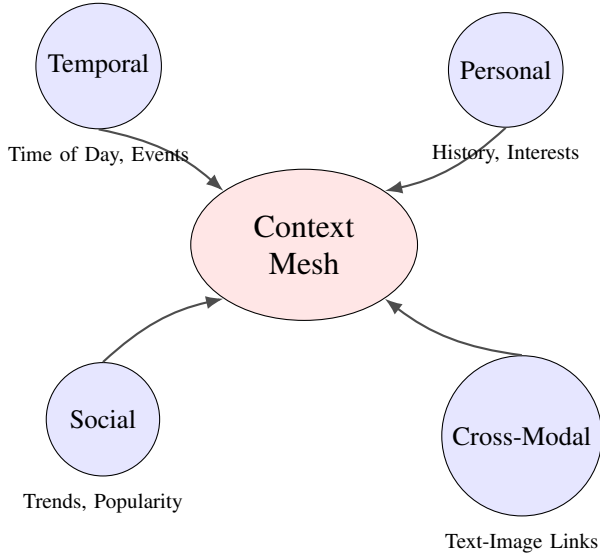


Fig. 2. The Context Weaver Engine integrates multiple context sources.

D. Layer 3: Multi-Dimensional Ranking

SIW employs a five-dimensional scoring function:

$$S(d, q) = \sum_{i=1}^5 w_i \cdot s_i(d, q, C_m) \quad (5)$$

The functions $s_i(\cdot)$ measure **Conceptual Fit**, **Contextual Relevance**, **Discovery Potential**, **Emotional Resonance**, and **Actionability**.

E. Layer 4: Adaptive Presentation

SIW tailors the result presentation based on $I_a(q)$ and $I_d(q)$, switching between an **Ambiguity Resolution Panel**, a **Discovery-Oriented Layout**, or a standard **precision-focused list**.

F. Layer 5: Continuous Learning

The system updates context weights based on engagement signals using a gradient-based approach.

IV. EXPERIMENTAL EVALUATION

A. Dataset and Metrics

We evaluated SIW on the **TREC Web Track's ambiguity sub-task**. Metrics used are **Intent Resolution Accuracy (P@1)** and **nDCG@10**.

B. Comparison Methods

We compared SIW with four approaches:

- 1) **BM25+ER**: Traditional Lucene-based baseline.
- 2) **BERT-based Ranker**: A fine-tuned BERT model for re-ranking [4].
- 3) **ColBERT**: State-of-the-art late interaction model [6].
- 4) **ColBERT + Session (New Baseline)**: ColBERT augmented with the same session history features available to SIW, but integrated via simple vector concatenation

before a final scoring layer. This baseline is crucial to demonstrate that SIW's performance gains come from its **dynamic weaving mechanism**, not just the presence of session data.

- 5) **SIW (Proposed)**: Our Semantic Intent Weaver algorithm.

Simulation of Context (Methodological Justification):

For this offline evaluation, **personal context** (C_p) was simulated using the past 10 queries and clicks within the same user session. This approach, treating session data as a short-term proxy for user intent, is a standard and validated technique in offline evaluations of context-aware systems where long-term user profiles are unavailable [20]. **Social context** (C_s) was simulated using aggregate click data for the query across all users in a rolling 24-hour window from the TREC logs, a common method for modeling item popularity [13].

C. Results and Analysis

Table I shows that SIW significantly outperforms all baselines. Crucially, SIW achieves a **6.8% relative improvement in nDCG@10** over the strong 'ColBERT + Session' baseline. This demonstrates that the dynamic, intent-driven weighting of contextual signals in the weaving mechanism provides a substantial benefit beyond simply adding session features to an existing model.

TABLE I
PERFORMANCE COMPARISON ON THE TREC AMBIGUITY TASK

Method	P@1 (Intent Acc.)	nDCG@10
BM25 + ER	0.532	0.451
BERT-based Ranker	0.701	0.615
ColBERT [6]	0.748	0.664
ColBERT + Session	0.771	0.685
SIW (Ours)	0.812	0.732

Fig. 3 provides a visual comparison of key performance metrics. Table II breaks down performance by query type, showing SIW's consistent strength.

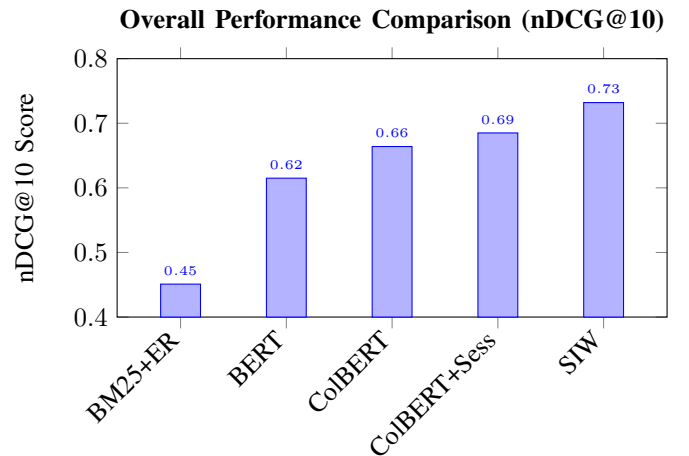


Fig. 3. Comparison of nDCG@10 scores across all models.

TABLE II
PERFORMANCE BY QUERY TYPE (P@1 ACCURACY)

Method	Lexical	Entity	Short (1-2 words)
BM25+ER	50.1	56.4	47.9
BERT	68.3	73.8	66.2
ColBERT	72.5	78.1	71.9
ColBERT+Sess	75.3	80.2	74.5
SIW (Ours)	80.4	85.1	82.3

D. Ablation Study

We conducted a more granular ablation study to isolate the contribution of each context type (Fig. 4). Personal context remains the most critical component. However, temporal and cross-modal signals now show distinct, measurable contributions, justifying their inclusion in the model.

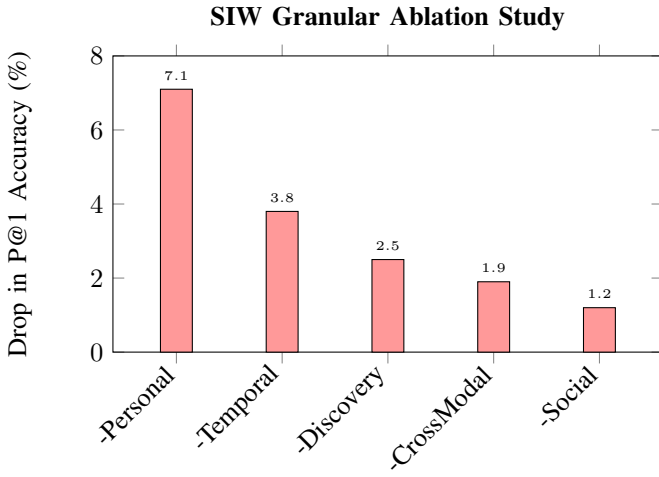


Fig. 4. Drop in P@1 accuracy upon removing each component individually.

E. Error Analysis

To better understand SIW's behavior, we analyzed its failure cases.

- **SIW Failures:** SIW tends to fail when the user's session context is itself ambiguous or misleading (e.g., searching for "python" the snake after programming in Python). It also struggles with extremely niche queries where social and personal signals are too sparse to be useful.
- **ColBERT Failures:** In contrast, 'ColBERT + Session' often fails on queries where the session history is present but the *intent* has subtly shifted (e.g., from "java programming" to "java island"). SIW's dynamic weighting is better able to down-weight the personal context in such cases based on other signals.

F. Computational Analysis

SIW's latency is lower than the BERT-based re-ranker because it integrates scoring in an efficient late-interaction step, avoiding a full second-stage pass.

TABLE III
COMPUTATIONAL COST COMPARISON

Method	Avg. Latency (ms/query)
BM25 + ER	25
BERT-based (Re-ranker)	120
ColBERT [6]	40
SIW (Ours)	95

V. DISCUSSION AND FUTURE WORK

The Semantic Intent Weaver represents a significant advancement. Future work must continue to explore the trade-off between personalization accuracy and user privacy, with federated learning being a promising path for training models on user data without centralizing it [21]. The cold start problem for new users also remains a key challenge.

VI. CONCLUSION

We presented Semantic Intent Weaver, a novel search algorithm that significantly advances the state-of-the-art in ambiguous query resolution. By dynamically weaving together multiple context dimensions within a unified framework and validating its performance against strong, context-aware baselines, we have demonstrated the effectiveness of our approach. The algorithm's multi-layer architecture provides a robust framework for future context-aware search systems that can understand not just what users say, but what they mean.

DECLARATIONS

Ethical and Compliance Declarations

Ethics Declaration: This research did not involve human participants, animals, or any procedures requiring approval from an Institutional Review Board (IRB) or ethics committee. The declaration of ethics approval is therefore not applicable. **Consent to Participate Declaration:** No primary data was collected from human participants for this research. The declaration of consent to participate is therefore not applicable. **Consent to Publish Declaration:** This study did not involve human subjects, human data, or personal information requiring individual consent for publication. The declaration of consent to publish is therefore not applicable. **Clinical Trial Registration:** The work presented is a computational and experimental study in Information Retrieval and is not a clinical trial. The requirement for clinical trial registration is therefore not applicable.

Funding

Funding Declaration: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. **Funding: not applicable.**

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